

## Project Design Phase Solution Architecture

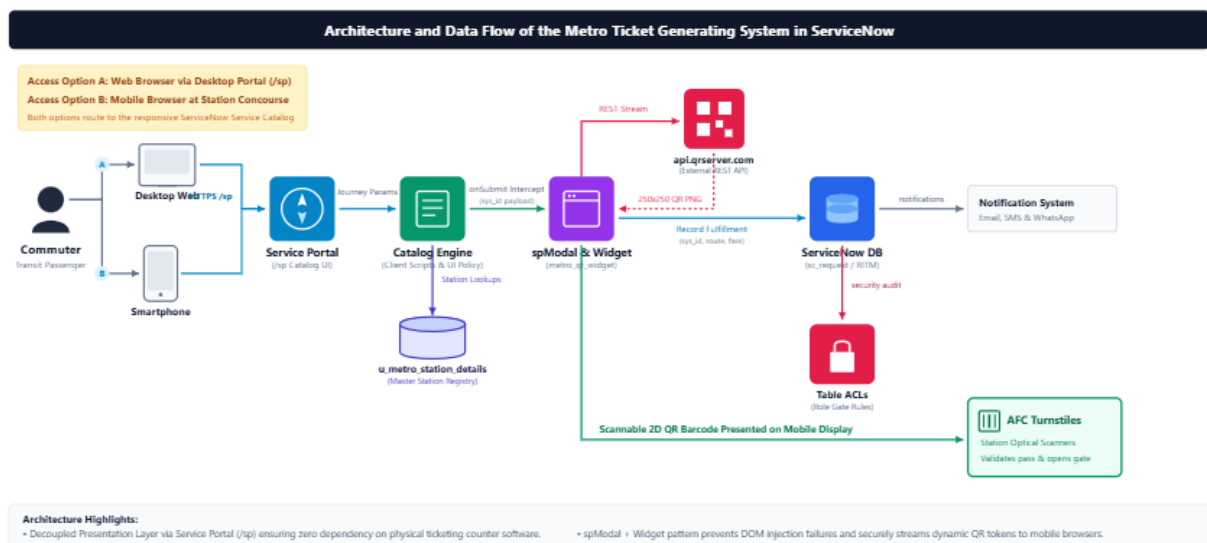
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|---------------|----------------------------------------------|
| Date          | 11 September 2026                            |
| Team ID       | NA                                           |
| Project Name  | Metro Ticket Generating System in ServiceNow |
| Maximum Marks | 4 Marks                                      |

### Solution Architecture Overview

The solution architecture bridges commuter needs with ServiceNow platform capabilities by establishing an automated, contactless transit ticketing flow:

1. **Client & Presentation Tier:** Commuters access the responsive Service Portal (/sp) using desktop or mobile browsers. The catalog item interface (Book A Metro Ticket) accepts transit route parameters without requiring local client installations.
2. **Dynamic Scripting & Logic Tier:** Client-side form interactions evaluate origin and destination stations using an onChange Catalog Client Script. This script calculates Single (type\_of\_journey = 1) and Return (type\_of\_journey = 2) fares in real time based on station coordinates. A Catalog UI Policy manages field visibility, displaying and enforcing the enter\_payment\_mode variable when "Others" is selected.
3. **Checkout Interception & Widget Rendering Tier:** Upon form submission, an onSubmit script intercepts the checkout event and constructs an encoded payload containing the unique record identifier (sys\_id). The script invokes an spModal dialog housing a custom widget (metro\_qr\_widget), which interfaces with a dynamic QR generation API to present a 250x250 scannable 2D barcode directly on screen.
4. **Data Management & Security Tier:** Station reference lookups are retrieved from the custom table u\_metro\_station\_details. Confirmed bookings trigger the creation of standard ServiceNow request records (sc\_request and sc\_req\_item). Table-level Access Control Lists (ACLs) permit public read access for station selection while restricting record manipulation to administrative roles.
5. **Physical Turnstile Integration Tier:** Station Automated Fare Collection (AFC) turnstiles equipped with optical barcode scanners read the mobile QR pass, validating transit authorization against the booking token.

### Solution Architecture Diagram



## Architectural Component Mapping

| Architectural Component   | Implementation Layer     | Operational Description                                                                                    | Platform Mechanism                                       |
|---------------------------|--------------------------|------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| Commuter Access Interface | Presentation Tier        | Enables commuters to access the ticket booking item on web or mobile devices.                              | Service Portal (/sp), Catalog Item (Book A Metro Ticket) |
| Dynamic Fare Processor    | Business Logic Tier      | Evaluates station origin-destination sys_ids and multiplies passenger counts against route matrix rates.   | Catalog Client Script (onChange on no_of_passengers)     |
| Payment Mode Validator    | Business Logic Tier      | Evaluates payment mode choices and toggles supplementary input variables when required.                    | Catalog UI Policy (Mode of Payment = Others)             |
| Token Interceptor Engine  | Orchestration Tier       | Intercepts the checkout submission, passes the record sys_id, and renders the modal ticket display.        | Catalog Client Script (onSubmit), spModal API            |
| Dynamic QR Generator      | Integration Tier         | External REST API that receives encoded ticket URLs and streams high-contrast 250x250 PNG barcodes.        | api.qrserver.com REST Integration                        |
| Station Master Store      | Relational Data Tier     | Custom table storing active transit stations, unique abbreviations, and routing identifiers.               | Custom Database Table (u_metro_station_details)          |
| Transactional Order Store | Relational Data Tier     | Enterprise database tracking passenger requests, fulfillment logs, and variable state entries.             | Core Schema Tables (sc_request, sc_req_item)             |
| Security & Access Layer   | Security Tier            | Enforces read availability for commuters while restricting schema modifications to administrators.         | Contextual Security & Role-Based ACLs                    |
| AFC Turnstile Validation  | Edge Infrastructure Tier | Optical gate scanners at metro entry points that read 2D barcodes from mobile screens to clear passengers. | Automated Fare Collection (AFC) Gate Scanners            |